

Project ESCAPE: Entropy & Semantic Compute Allocation in Programming Environments: Structural Discovery via Entropy in the Byte Latent Transformer Across Code and Natural Language

Team Trimax · CS7.501 Advanced NLP · Final Project Proposal

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Abstract

Most language models split text using a fixed vocabulary learned before training, such as byte-pair encoding. Every input is then chopped the same way, whether it is boilerplate or dense logic. The Byte Latent Transformer (BLT) (Pagnoni et al., 2024) does something different: it reads raw bytes and decides where to split *as it goes*, cutting whenever its own next-byte prediction becomes uncertain. Those cut points are the model reporting where its knowledge ran out.

Nobody has checked whether they mean anything. We ask a narrow, falsifiable version of the question: **do BLT’s boundaries land at the start of syntactic constituents in code?** We call this the Grammatical Branching Hypothesis. We give a mechanism for why it should hold, state plainly where that mechanism must fail, and test it against a null model that holds boundary density fixed, so a positive result cannot be dismissed as an artifact of BLT’s threshold τ . Prose is a built-in negative control: if code and prose do not separate, we are measuring τ , not grammar. We further test whether an entropy-only mechanism can be mistaken for semantic risk-awareness in memory-unsafe C++, and test whether the two can be told apart. The protocol is designed so that a null result is as informative as a positive one.

1 Introduction

Tokenization has been the standard front end for language models for a decade, but it fixes a compression ratio in advance: a tokenizer trained before the model sees your data spends the same number of tokens on boilerplate as on subtle logic. BLT removes that fixed vocabulary. A small byte-level language model, the *patcher*, reads raw bytes and, at each position, measures how uncertain it is about the next byte via Shannon entropy:

$$H(b_i) = - \sum_v p(v | b_{<i}) \log p(v | b_{<i}) \quad (1)$$

When H crosses a threshold τ , BLT starts a new *patch*, so a large transformer spends more compute where bytes were hard to predict and less where they were easy.

A boundary is therefore not a preprocessing decision but **a claim the model makes about its own uncertainty**, interpretable in a way a BPE merge table never is, since we can

ask whether the claim corresponds to anything independently verifiable. The BLT authors demonstrate the approach works, up to 8B parameters on 4T training bytes, but report only efficiency and downstream accuracy; they never check whether the induced patches line up with structure we can verify. That is the gap this project addresses, and code is the place to look: it has machine-checkable ground truth in its abstract syntax tree (AST), with grammar choice points that flexible prose does not have.

2 Motivation and Related Work

Interpretability work on tokenization has focused almost entirely on *static* vocabularies. Byte-pair encoding (Sennrich et al., 2016) fixes a merge table before training, with well-documented failure modes: poor morphological alignment, unstable segmentation of rare or multilingual tokens. Tokenization-free models remove the vocabulary but not the uniformity: CANINE (Clark et al., 2022) and ByT5 (Xue et al., 2022) process every byte with equal compute, never deciding *where* a boundary belongs, and BLT is the first architecture at scale where segmentation is both byte-level and dynamic.

Structural probing work, including CodeBERT (Feng et al., 2020) on code and syntax probes on contextual representations (Hewitt and Manning, 2019), shows *learned representations* can encode syntax, but only because the model was trained or probed *for* that structure; none asks whether an unsupervised, compression-only signal with no parser in the loop recovers syntax as a side effect.

Two gaps follow: no work characterizes whether BLT’s patcher implicitly recovers constituent structure, and none establishes whether any such recovery is genuine grammar-tracking or merely an artifact of patch density pinned by τ : a distinction that requires a principled null model, not a raw overlap score.

3 Contributions

1. **A stated mechanism, not a black-box metric.** A specific, falsifiable account (§5.2) of *why* boundaries should fall at constituent starts, including explicit statement of where it must fail.
2. **A null model that makes the result mean something.** A density-matched baseline plus a whitespace baseline (§5.3), so measured alignment is attributable to grammar rather than to boundary density or to newlines.
3. **A built-in negative control.** Prose is not a side experiment: the same mechanism predicts a far weaker effect there, so failure to separate code from prose is itself evidence we are measuring τ .
4. **An honest scope limit.** We test what an entropy-only mechanism *cannot* see (§5.5), so we do not overclaim that surface predictability implies danger-awareness.

4 Research Objectives

- 01. Syntactic Alignment.** Do BLT’s entropy-triggered boundaries align with AST node boundaries in C++ and Python, beyond a density-matched baseline?
- 02. Compute Allocation Ratio.** How do average Bytes-Per-Patch (BPP) and its variance differ between rigid code syntax and natural language?
- 03. Cross-Language Structural Recognition.** Does patching behave differently for indentation-based syntax (Python) versus bracket-based (C++)?
- 04. Robustness to Code Noise.** Do boundaries survive character-level noise (syntax errors, obfuscation), or does the mechanism fail systemically?
- 05. Semantic vs. Surface Compute Allocation.** Is any extra compute BLT gives to memory-unsafe C++ operations *fully explained* by the surface unpredictability of nearby identifiers and casts, or does a residual remain after conditioning on that confound?

5 Hypothesis

5.1 The Grammatical Branching Hypothesis (H1)

BLT’s patch boundaries are not random compression artifacts. They concentrate at the **start** of syntactic constituents rather than inside them; this concentration is stronger in code than in prose, and stronger at constituent starts than at their ends.

Stated precisely, H1 claims an *average-case asymmetry* between constituent starts and constituent interiors. It does **not** claim that constituent starts are points of maximal grammatical choice. §5.2 explains why that stronger claim is false and why it is not needed. The project measures how large the asymmetry actually is.

5.2 Why We Expect This to Hold

BLT’s patcher is a small byte-level language model with a short (512-byte) context window, emitting a boundary whenever next-byte entropy crosses τ . A boundary therefore marks exactly one thing: **a position where the preceding context stops determining what comes next**. Everything below follows from that sentence, and we are careful to claim no more than it supports.

Entropy bounds branching; it does not track it. It is tempting to say entropy measures the local branching factor B , the number of grammatically valid continuations. It does not: entropy depends on the *shape* of the distribution, not the size of its support, and the only guaranteed relation is $H \leq \log B$. Ten valid continuations with one carrying 99% of the mass give $H = 0.08$ nats, well below $\tau = 1.34$; four equiprobable continuations give $H = 1.39$ nats and cross it. Branching is necessary for high entropy but nowhere near sufficient.

What survives is the bound in one direction. A boundary *certifies* the local distribution was genuinely flat: $H > \tau$ implies $B > e^\tau \approx 3.8$, so every entropy-triggered boundary sits where at least four continuations were live: a **subset** of high-branching positions, where the grammar has not yet concentrated its mass on one continuation. This yields a prediction we state in advance: **precision against constituent starts should exceed recall**.

The load-bearing claim is about interiors, not starts. Our argument does not require constituent starts to be maximal-choice points, only that entropy be lower *inside* a

Position	Entropy	Split?	Case
i of if	high	yes	correct
(after if	low	no	<i>missed</i> start
x of x > 0	high	yes	lexical, not syntactic
) ; }	low	no	correct (supports P2)

Table 1: The mechanism applied to `if (x > 0) { total += 1; }`. Row 1 is the effect we predict. Row 2 costs recall (forced opener). Row 3 costs precision (identifier entropy). Row 4 is why starts should beat ends.

committed production than at its boundary: a structural property of context-free grammars, not a statistical hope. Once a production is entered, admissible continuations are constrained by the production itself: mid-keyword and mid-identifier bytes are cheap, and closing delimiters (`;`, `)`, `}`) are among the most predictable bytes in a file, since the grammar requires them once committed rather than choosing them. This is the asymmetry P2 rests on: boundaries are pushed *away from* interiors and right edges, and constituent starts inherit them by elimination, not by being choice-maximal.

Where this mechanism does not hold. Two dissociations follow directly, because they hit *different* metrics. First, some constituents open deterministically: in both C++ and Python, `if`, `for` and `while` are followed by a forced token, and `def` must be followed by a name. These starts have branching factor near one, so the mechanism predicts BLT will miss them, costing **recall**, tested directly via P4. Second, high entropy also arises where no constituent opens at all: partway through an identifier, inside a string literal or comment. These are lexical choice points, not grammatical ones, costing **precision**, the confound isolated in §5.5. The patcher measures next-byte predictability, conflating grammatical branching with lexical arbitrariness; nothing in the architecture separates them.

Table 1 walks one line of C++ through all three cases.

Prose is the contrast case. Branching factor is flatter across a sentence (content words are hard everywhere, function words easy everywhere, and no formal rule forces a clause to end with a specific character), so prose should show more uniform patch lengths and weaker structural anchoring. **This code/prose gap is what makes the claim one about grammar rather than about τ .**

5.3 Predictions and Null Model

P1: Alignment. Boundary positions match AST node start offsets, within a tolerance k fixed *a priori* per language on a held-out calibration split, well above baseline, for statement- and expression-level nodes.

P2: Asymmetry. Alignment at node starts is clearly higher than at node ends. Closing delimiters are low-entropy, so BLT should systematically miss right edges. Span-IoU therefore *understates* the effect; boundary precision/recall against start offsets, not IoU, is our primary metric.

P3: Cross-domain contrast. BPP variance is high in code and low in prose, and P1-style alignment in prose (against a constituency parse) is near chance. Code and prose must dissociate.

P4: Stratification by opener. Alignment is measurably weaker for constituent types with deterministic openers (`if_statement`, `for_statement`) than for open-ended

ones (`expression_statement`, `return_statement`, assignment right-hand sides). P4 follows directly from §5.2 and is the sharpest test that the mechanism running is the one we claim: a generic “entropy correlates with structure” story does not predict it.

Null model (H0). Agreement between patch boundaries and AST node starts is no better than a density-matched random boundary set: the same number of boundaries, patch lengths drawn from BLT’s own empirical patch-length distribution and laid out in shuffled order, truncating the final sampled patch so the boundary set exactly spans the file.

This baseline is required, not optional: mean patch length is roughly pinned by τ , so *any* boundary set of the right density scores nonzero raw alignment, and only the margin over H0 is informative. We assess it with a permutation test (10,000 resamples), reporting effect size with confidence intervals rather than a raw overlap number.

Second baseline: whitespace. H0 is structure-blind by construction, so it cannot rule out the most obvious confound in code: most AST node starts sit immediately after a newline and indentation, so a trivial “split at every newline” rule would score well on P1. H1 is only interesting if BLT beats *that*, not merely chance, which matters most in Python, where indentation *is* the grammar. Alignment is additionally computed *per AST depth*, since every byte sits inside several nested nodes and “alignment” is undefined until granularity is fixed.

What counts as a result. H1 is supported if P1 clears both baselines with non-overlapping confidence intervals, P2 holds in the predicted direction, P3 shows code/prose dissociation, and P4’s stratification appears. Any subset failing is reported as such; §5.4 explains why each failure mode is itself interpretable.

5.4 Why the Result Is Not Obvious in Advance

If H1 holds, an unsupervised, entropy-driven compressor recovers syntactic constituent boundaries with no parser in the loop and no code-specific tokenizer. If H1 fails, *how* it fails is itself the signal: uniformly **high** boundary entropy would mean the patcher is undertrained on code and allocates compute blind to grammar (fixable by retraining on code); uniformly **low** entropy would mean code is so compressible that τ rarely fires, so boundaries are density-limited rather than structure-driven. The sharpest way H1 could lose outright is identifier entropy dominating and swamping the syntactic signal, the precision failure of §5.2, addressed directly in §5.5. Because each failure mode is diagnosable rather than a single pass/fail outcome, the experiment is informative regardless of which way the result goes.

5.5 Scope Limit: What Entropy Cannot See

The patcher is a short-context, surface-statistics model with no representation of pointer aliasing, ownership, or memory safety. Tokens like `new`, `delete`, `*` and `&` are short, frequent and lexically predictable, so the mechanism of §5.2 predicts **low** entropy at them, not high.

Consequently **we do not expect BLT to allocate compute by semantic risk**. That claim would directly contradict the mechanism our own hypothesis relies on. Any extra compute observed in memory-unsafe regions should instead come from the arbitrary identifiers and casts that tend to surround pointer operations.

Objective 5 tests this directly: we condition memory-specific compute profiling on identifier-token entropy via regression, and predict that any apparent memory-safety effect shrinks toward zero once that confound is controlled. This is a confound-isolation test, not a search for semantic understanding. A null result here is informative, not a failed experiment.

6 Methodology

Figure 1 shows the pipeline end to end.

Model and data. We use `BlTForCausalLM` from the Hugging Face `transformers` library, specifically the `itazap/blt-1b-hf` checkpoint (a 1B-parameter port of BLT’s Local Encoder $\rightarrow$ Latent Transformer $\rightarrow$ Local Decoder architecture, and the largest openly available BLT checkpoint at present). Natural language is sampled from WikiText-103, Python code from CodeSearchNet, and C++ from The Stack (`.cpp/.hpp` files; CodeSearchNet has no C++ split). For the prose negative control (P3) we additionally parse WikiText-103 sentences with a standard constituency parser to obtain phrase-boundary ground truth.

Boundary extraction. BLT resolves patches dynamically from the entropy threshold ($\tau \approx 1.34$) and an optional maximum patch length; the patcher exposes per-byte entropies and patch lengths directly, so boundaries are read from its output rather than by instrumenting internals. Each boundary is tagged *entropy-triggered* or *length-capped*, since a patch hitting the length cap before entropy crosses τ has nothing to do with grammar, and length-capped boundaries are excluded from the primary P1/P2 scores. Entropy is reported in nats, so $\tau = 1.34 \text{ nats} = 1.93 \text{ bits}$.

Checkpoint fidelity. The only public checkpoint is a third-party conversion, so we sanity-check it before producing any numbers: coherent generation, sensible boundaries on simple inputs, and, critically, that the effective context window and patch-length distribution match those reported in the original work, since the short window is load-bearing for §5.2 and a silent mismatch would invalidate the mechanism we argue from.

AST ground truth. We parse code with `tree-sitter` for byte-level start/end offsets of statement-level nodes (`if_statement`, `for_statement`, `declaration`) and expression-level nodes (`call_expression`, `binary_expression`). Separately we query pointer dereferencing (`*`), address referencing (`&`) and manual memory management (`new/delete`) as memory-unsafe regions.

Adversarial data. We build a noised variant of the code corpus by injecting missing semicolons, localized indentation changes, camelCase $\leftrightarrow$ snake_case inversions, and typos in syntax keywords, to probe robustness (O4). Noise breaks token patterns without destroying logical validity; mutated files are re-parsed with `tree-sitter` and any that fail to produce a clean tree are discarded, so degraded alignment cannot be blamed on degraded ground truth.

7 Evaluation Metrics

- **Boundary P/R at constituent starts (primary), overall and stratified by node type.** Precision and recall of entropy-triggered boundaries against AST node start offsets within tolerance k , as a margin over *both* baselines,

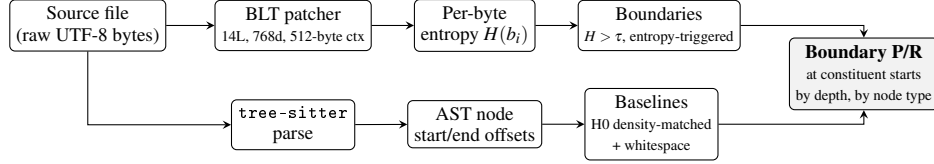


Figure 1: Extraction and scoring pipeline. The upper path is unsupervised: the patcher never sees a parse tree. The lower path supplies independently verifiable ground truth. Boundaries are scored against AST node starts and against both baselines simultaneously, stratified by AST depth (§5.3) and by node type (P4).

and separately grouped by constituent type (deterministic vs. open-ended openers, the test for P4). Where node types share a start offset, a boundary is credited once, against the outermost matching node, so nested starts cannot inflate precision. We expect precision > recall (§5.2).

- **AST boundary IoU (secondary).** Byte-span overlap between patches and AST nodes; expected to understate the effect per P2, reported for comparability with prior segmentation work.
- **Compute density (BPP) and entropy heatmaps.** Bytes-per-patch profiling and entropy visualizations over source text; we expect high BPP variance for code versus more uniform BPP for prose.
- **Degradation shift.** Change in the primary metric and in BPP between clean and adversarial data, indicating robustness relative to subword tokenizers.
- **Memory-specific compute profiling.** BPP in memory-unsafe regions before and after conditioning on identifier-token entropy, isolating a genuine semantic-risk effect (if any) from the surface-predictability confound.

8 Feasibility and Compute

The study is deliberately cheap: everything H1 requires comes from the *patcher* (roughly 150M parameters; see Figure 1), never the 1B latent transformer. Inference in `bf16` fits on a single consumer GPU as a forward pass with no training, so the experiment is reproducible on modest hardware; larger corpus sweeps and the adversarial condition (O4) can dispatch as institutional batch jobs if throughput becomes the limit. **No model training is required at any stage.**

9 Risks and Mitigations

Risk	Mitigation
Checkpoint infidelity	Verified against the original configuration before any results are produced (§6); gates everything downstream.
Newline confound	Explicit whitespace baseline (§5.3); H1 must beat it, not merely H0.
Identifier entropy swamps syntax	Predicted in advance (§5.2), isolated by regression in O5; recall loss is total but confined to few node types, so precision should still net dominate.
Mechanism argued from bracket syntax	Direct for C++’s closing delimiters; Python dedents instead, so O3, not theory, decides whether interiors stay cheap there.
Parser offset drift	Character offsets round-tripped to UTF-8 byte offsets and validated before scoring.
Depth incomparable across languages	Alignment computed per depth and normalized before cross-language comparison (O3).

Table 2: Principal risks and how each is handled. The first is a precondition, resolved in Weeks 1–3 before any alignment number is generated.

10 Deliverables

The boundary-extraction pipeline (§6), the *tree-sitter* mapping engine, and the evaluation harness implementing P1–P4 and both baselines (§7) are each runnable independently. The project’s deliverable is a final report and presentation covering all five objectives, including negative results.

11 Timeline

Weeks	Work
1–3	Extraction pipeline; <i>tree-sitter</i> AST mapping for Python/C++; checkpoint and context-window sanity checks.
4–6	H0 null model and whitespace baseline; P1/P2 alignment on code; P4 stratification; prose contrast (P3).
7–8	Mid-submission write-up; adversarial dataset; O4.
9–11	Memory-unsafe profiling and confound conditioning (O5); cross-language comparison (O3).
12+	Consolidation, ablations, final write-up and presentation.

Table 3: Tentative timeline.

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